

IBM Bob Developer Onboarding Copilot

Transform any complex GitHub repository into a personalized, interactive onboarding experience powered by IBM watsonx & Bob.

Track: IBM Bob 2.0 Hackathon (AI-Assisted Development) | Tech: IBM watsonx.ai (LLaMA 3.3 70B), MCP, FastAPI, React

The Problem: Developer Onboarding is Slow & Expensive

Traditional onboarding wastes weeks of engineering time and context-switching budget

2 to 4 Weeks Lost

New engineers spend weeks deciphering repository structures, setup scripts, and business logic before making their first commit.

Context Switching Overhead

Senior engineers spend hours per week answering repetitive architecture and dependency questions.

Outdated Documentation

READMEs, wikis, and setup guides quickly fall out of sync with fast-moving codebases.

High Financial Friction

Onboarding delays cost engineering organizations thousands of dollars per developer in lost productivity.

The Solution: Your 24/7 AI-Powered Senior Mentor

Instant, intelligent repository onboarding from first clone to first pull request

1. Paste GitHub URL

Connect any public or private repository URL directly in the dashboard.

2. Automated Structural Parsing

IBM watsonx.ai parses entry points, dependencies, and codebase structure.

3. Personalized Roadmaps

Generates role-adapted (Junior/Mid/Senior) day-by-day onboarding plans.

4. Context-Aware Q&A

Answers codebase architecture questions with line-level citations.

5. Pre-Merge Code Validation

Automatically tests and validates local task code against acceptance criteria.

6. 1-Click PR Generator

Curates first issues and auto-generates pull request titles & descriptions.

Core Features: From First Clone to First PR

Engineered for maximum developer velocity and seamless AI assistance



Personalized Onboarding Plan

Dynamic multi-day learning modules complete with actionable checklists, estimated completion times, and file references.



Codebase Chat (LLaMA 3.3 70B)

Context-aware conversational assistant hosted on IBM Cloud watsonx.ai providing line-specific codebase answers.



Interactive Code Validation

Real-time pre-merge AI evaluation of local code submissions against task acceptance criteria.



PR Title & Description Generator

Auto-curates beginner-friendly 'good first issues' and formats standardized GitHub PR pull request artifacts.

System Architecture & Tech Stack

Powered by IBM watsonx.ai, Model Context Protocol (MCP), FastAPI & React

Architecture Diagram

```
[ React + TypeScript Frontend ]  
  | Interactive Onboarding Dashboard & Chat UI  
  ▼  
[ FastAPI Model Context Protocol (MCP) Server ]  
  | Repo Parsing Engine & Tool Declarations (/mcp/)  
  ▼ (IAM OAuth2 Token Exchange)  
[ IBM watsonx.ai Foundation Models ]  
  └─ meta-llama/llama-3-3-70b-instruct
```

Tech Stack Summary

- AI Engine: IBM watsonx.ai
- Model: LLaMA 3.3 70B
- Protocol: Model Context Protocol (MCP)
- Backend: Python, FastAPI
- Frontend: React, TypeScript, TailwindCSS
- Deployment: Vercel & Docker

Impact & Future Vision

Accelerating developer productivity and team velocity



Key Hackathon Outcomes & Future Roadmap

- ⚡ 70% Reduction in Developer Ramp-up Time: From weeks to hours from first repo clone.
- 🤖 Native IBM Bob & watsonx Orchestrate Integration: Standard MCP tool exposure for AI agents.
- 📊 Enterprise Telemetry: Team progress monitoring, skill gap analysis, and onboarding metrics.
- 🌐 Open Source & Production Ready: Configured for Vercel deployment with full test coverage.